

How Does J-Space Ablation Trade Off Against External CoT?

CS 2881r (AI Safety) — Homework 0

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1. Hypothesis

The workspace paper identifies a small set of verbalizable directions (the *J-space*) that carries a model's intermediate reasoning state, and shows that ablating it devastates multi-hop reasoning while leaving shallow extraction intact. Chain of thought externalizes intermediate state into the context, where an activation-space ablation cannot reach it. Our pre-registered hypothesis (recorded in `src/config.py` before any ablated data was generated) is a substitution account:

H1 (substitution). J-space ablation hurts *direct* answering (no chain of thought) more than *CoT* answering, because CoT can re-externalize the work the workspace was doing. The pre-registered headline statistic is the interaction

$$\Delta = (\text{direct_intact} - \text{direct_ablated}) - (\text{cot_intact} - \text{cot_ablated}),$$

with H1 predicting $\Delta > 0$. A $\Delta \approx 0$ with a tight interval means no measurable trade-off; $\Delta < 0$ would indicate the opposite of substitution — that the workspace remains load-bearing *during* chain of thought. Both outcomes are visible under the same design, which is what makes the test fair.

H2 (difficulty scaling). As problem difficulty rises (GSM8K $\rightarrow$ MATH-500 $\rightarrow$ AIME 2024), substitution predicts the CoT arm stays robust to ablation even as problems harden, since more of the computation is externalized. The complementary account predicts the ablation deficit in the CoT arm grows with difficulty, because harder problems make the model lean on internal workspace state between emitted tokens.

Selectivity requirement. Under either hypothesis, a deficit only counts as a J-space effect if it exceeds a matched random-direction control; a deficit shared with the control is broad degradation, not workspace removal.

2. Experiment design

Per dataset, a 2x3 factorial: an *externalization axis* (cot: thinking mode enabled; direct: thinking disabled, an “answer only, no reasoning” instruction, and a `\boxed{` prefill that commits the model to answering immediately) crossed with an *intervention axis* (intact / J-space ablated / random-direction control). All six cells run the same problems, so every comparison is paired at the problem level.

Controls that separate J-space effects from broad degradation

- **Matched random control (rand_tok).** The identical operation — project out $k = 10$ directions drawn from the unembedding, in the same layer band, with the same projection mode — with only the selection randomized: k uniformly random unembedding rows instead of the top- k readout tokens, re-drawn per (problem, layer, position). Anything the ablation does that the control also does is degradation, not a workspace effect.
- **Control interaction.** The full interaction Δ is recomputed with the random cells substituted for the ablated ones. It should be ≈ 0 ; if it is not, the pipeline itself (not J-space) moves the headline number, and the two must be read together.
- **Selectivity per arm.** $(\text{intact} - \text{ablated}) - (\text{intact} - \text{random}) = \text{random} - \text{ablated}$, with a paired bootstrap CI: the targeted-minus-matched-control deficit within each arm.

- **Degeneration gate.** Before paying for the expensive cells, ablated CoT runs on the first 20 problems and the run stops if its unusable rate (incomplete / unparsed / error) exceeds the matched intact cell's by more than 15 points — a tripwire against comparing arms when generation has simply broken. This gate fired on AIME (§4).

Statistics and pre-registration

Within-arm comparisons use McNemar's exact test on paired outcomes; selectivity and interaction estimates use a percentile bootstrap (10,000 resamples) over *problems*, not observations, preserving the pairing. Sample sizes came from a power analysis at pilot accuracies ($\rho = 0.5$): $n = 150$ for GSM8K; $n = 150$ for MATH-500 (amended from 100 before any MATH-500 data existed); all 30 problems for AIME 2024. Every analysis choice — layer band, k , projection mode, exclusion rule, token caps, sample sizes, prompts — was fixed in a committed config module before ablated data was generated, and accessors raise on any undecided value so a choice cannot be made silently at the keyboard. Twelve GSM8K problems were spent selecting the layer band; the analysis flags them and reports the exploratory-vs-holdout split alongside the headline numbers, since selection on outcome data inflates whatever it selected for.

Why this design answers the question: substitution, complementarity, and no-effect each produce a distinct, visible signature — positive, negative, or null interaction — and the matched control plus control interaction ensure that whatever signature appears cannot be explained by the intervention simply damaging the model.

3. Experimental details

Model and decoding. Qwen3-4B at a pinned checkpoint revision, bf16, greedy decoding, batch size 1, single device per run. Greedy bf16 decoding is deterministic only within a backend (MPS vs CUDA kernels differ), so the device is recorded on every record and a run refuses to resume onto a different backend.

J-lens under the feasibility approximation $J = I$. The paper's J-lens requires per-layer Jacobians averaged over positions and a 1,000-prompt corpus, which is out of reach at our compute. We use the special case $J = I$: the J-lens vector for token t at every layer is the unembedding row $W_U[t]$ scaled by the final RMSNorm's learned gain, and the readout is the gain-aware logit lens. Calibration against the model's own logits at the final layer: 100% top-1 agreement, 94.1% top-10 set agreement (the discrepancy is bf16 rounding at the rank-10 boundary; the full norm is applied to make the arithmetic exact). This is the single largest deviation from the paper and is discussed in §5.

Ablation. At every position in the band, the top $k = 10$ vocabulary tokens by gain-aware readout are selected and their J-lens directions projected out of the residual stream (“each” mode — sequential per-direction projection, the paper's literal wording; a directions probe measured the selected ten as near-orthogonal, effective rank 9.13/10, so “each” and “span” differ little). The band is the paper's *light* band L38–54 on its 0–100 fractional-depth scale, mapped to layers 14–19 of Qwen3-4B's 36. Light rather than heavy because the paper reports that on its smallest model ablation degraded coherence before producing any qualitative change, and Qwen3-4B is far smaller still.

Exclusion rule. Following the paper, J-lens vectors of tokens in the top-10 of a clean forward pass on the same context are exempted at each position, so the intervention targets internal state rather than the readout of the token being emitted. This requires an interleaved clean forward pass at every generation step ($\approx 2\times$ cost on ablated cells). It is not optional: with the rule off, a damage-floor probe showed an 84% flip rate on SST-2 collapsing to 8% with it on — the rule-off “effect” was destruction of the answer readout itself.

Datasets and sampling. GSM8K test split, $n = 150$ by seeded shuffle prefix (samples nest across pilot and full runs by construction). AIME 2024, all 30 problems. MATH-500, $n = 150$ stratified 30 per difficulty level — note

this makes cell accuracies level-balanced averages, not comparable to published split-weighted MATH-500 numbers. The layer band was selected on 12 GSM8K problems and transferred unchanged to the other datasets; on those datasets the transfer is an assumption, not a measurement.

Token caps. Per dataset and condition, calibrated on intact generations where possible: GSM8K cot 3072 / direct 128; MATH-500 cot 16,384 / direct 128 (a disclosed budget cap — $\approx 15\%$ of level-5 intact CoT generations still hit it, per the stopping rule committed before the measurement was read); AIME cot 32,768 / direct 512 (an uncalibrated budget ceiling bounded by the checkpoint's 40,960-position limit). Generations that hit the cap score as “incomplete” (counted incorrect); compositions are reported per cell because a cap that binds asymmetrically would leak into the interaction.

Scoring. First `\boxed{}` match extracted and checked with a pinned math-verify; the direct condition's prefill has already opened the box, so trailing junk cannot change a score.

Compute limitations. Runs were generated on rented CUDA pods, never mixing backends within a comparison. The AIME cot_ablated cell holds 20 of 30 problems by protocol, not by budget: too many ablated generations ran to the 32,768-token cap, and the pre-registered degeneration gate fired and stopped the cell after its 20-problem window (§4). The MATH-500 cot_ablated cell ran to completion once, its records were accidentally lost before merging, and the cell was regenerated on 8/6 under the same pre-registered condition and cap. The regeneration necessarily happened after the other five cells' results were known; nothing about the condition is chosen at run time, but the ordering is disclosed.

4. Experimental results

GSM8K (n = 150)

Condition	n	Acc.	Correct	Incorrect	Incomplete
cot_intact	150	80.7%	121	5	24
cot_ablated	150	74.7%	112	8	30
cot_random	150	76.7%	115	7	28
direct_intact	150	27.3%	41	109	0
direct_ablated	150	24.7%	37	113	0
direct_random	150	24.7%	37	113	0

Table 1. GSM8K cell accuracies and outcome composition (n = 150 per cell).

Contrast	Point (pts)	95% CI (pts)	p	n pairs
CoT: ablated vs intact (McNemar)	−6.0	—	0.035	150
CoT: random vs intact (McNemar)	−4.0	—	0.180	150
Selectivity, CoT arm	+2.0	[−3.3, +7.3]	0.532	150
Selectivity, direct arm	+0.0	[−5.3, +5.3]	1.000	150
Interaction (ablation)	−3.3	[−10.7, +4.0]	0.407	150
Interaction (random control)	−1.3	[−7.3, +4.7]	0.678	150

Table 2. GSM8K paired contrasts. Points in percentage points; CIs are 10,000-resample paired percentile bootstraps over problems; McNemar p-values are exact.

Ablation lowered CoT accuracy by 6.0 points (McNemar $p = 0.035$), but the matched random control lowered it by 4.0 points ($p = 0.18$): selectivity is +2.0 points with a 95% CI of $[-3.3, +7.3]$. The direct arm shows identical 2.7-point drops under ablation and control (selectivity exactly 0). The headline interaction is -3.3 points, CI $[-10.7, +4.0]$ — the wrong sign for H1 and consistent with zero — and the control interaction is -1.3 points, CI $[-7.3, +4.7]$, confirming the pipeline itself does not manufacture an interaction. The 12 problems the band was selected on are reported apart from the 138-problem holdout: they are measurably unrepresentative (direct_intact 50% vs the holdout's 25%) and show no ablation deficit at all (83% in all three CoT cells vs 80/74/76% on the holdout), so the headline estimates — which include them as 12 of 150 pairs — are, if anything, diluted rather than inflated by the band-selection set.

MATH-500 (n = 150, stratified 30/level)

Condition	n	Acc.	Correct	Incorrect	Incomplete
cot_intact	150	94.0%	141	3	6
cot_ablated	150	88.0%	132	1	17
cot_random	150	96.0%	144	3	3
direct_intact	150	34.0%	51	99	0
direct_ablated	150	30.7%	46	104	0
direct_random	150	28.7%	43	107	0

Table 3. MATH-500 cell accuracies (level-balanced averages, not comparable to split-weighted published numbers). The cot_ablated records were accidentally lost after a completed run and the cell was regenerated on 8/6 — after the other five cells' results were known; the condition and cap are pre-registered and identical.

Contrast	Point (pts)	95% CI (pts)	p	n pairs
CoT: ablated vs intact (McNemar)	-6.0	—	0.078	150
CoT: random vs intact (McNemar)	+2.0	—	0.453	150
Direct: ablated vs intact (McNemar)	-3.3	—	0.227	150
Direct: random vs intact (McNemar)	-5.3	—	0.008	150
Selectivity, CoT arm	+8.0	[+3.3, +13.3]	0.001	150
Selectivity, direct arm	-2.0	[-6.7, +2.7]	0.487	150
Interaction (ablation)	-2.7	[-10.0, +4.7]	0.493	150
Interaction (random control)	+7.3	[+2.7, +12.7]	0.003	150

Table 4. MATH-500 paired contrasts. The CoT-arm selectivity is the only contrast in the study whose CI excludes zero.

Ablation left MATH-500 accuracy largely intact in absolute terms: the CoT arm drops 6.0 points to 88.0% (McNemar $p = 0.078$, marginal), and the drop is almost entirely non-termination rather than wrong answers — incorrect counts actually *fall* from 3 to 1 while incompletes rise from 6 to 17, all 17 of them hits on the 16,384-token cap. But the contrast against the matched control is sharp: the random control scores 2.0 points *above* intact, so CoT-arm selectivity is **+8.0 points, 95% CI [+3.3, +13.3], $p = 0.001$ — the only selectivity in the study whose CI excludes zero**. Which ten directions are removed matters in this arm. The direct arm shows the reverse: ablation costs 3.3 points (n.s.) but the random control costs 5.3 ($p = 0.008$, the only McNemar-significant drop), so direct-arm selectivity is $-2.0 [-6.7, +2.7]$ — null, consistent with GSM8K. The

ablation interaction is -2.7 $[-10.0, +4.7]$ (null, wrong sign for H1), and the control interaction is $+7.3$ $[+2.7, +12.7]$, $p = 0.003$ — **significantly non-zero**: removing ten *random* directions hurts no-CoT answering more than CoT answering here, so the ablation interaction must be read against $+7.3$, not zero — and its distance from that baseline, -10.0 points, is just the two selectivities' difference. Intact CoT incompletes are 6/150 (4%), consistent with the disclosed budget-cap expectation.

AIME 2024 (n = 30; cot_ablated n = 20)

Condition	n	Acc.	Correct	Incorrect	Incomplete
cot_intact	30	66.7%	20	5	5
cot_ablated	20*	55.0%	11	1	8
cot_random	30	73.3%	22	7	1
direct_intact	30	0.0%	0	30	0
direct_ablated	30	0.0%	0	30	0
direct_random	30	0.0%	0	30	0

Table 5. AIME 2024 cell accuracies. *cot_ablated stopped at 20 of 30 problems: the pre-registered degeneration gate fired on its cap-hit rate (see text); its paired contrasts use those 20 pairs.

Contrast	Point (pts)	95% CI (pts)	p	n pairs
CoT: ablated vs intact (McNemar)	-11.7	—	0.219	20
CoT: random vs intact (McNemar)	$+6.7$	—	1.000	30
Selectivity, CoT arm	$+20.0$	$[+0.0, +40.0]$	0.115	20
Selectivity, direct arm	$+0.0$	$[+0.0, +0.0]$	1.000	30
Interaction (ablation)	-20.0	$[-40.0, +0.0]$	0.113	20
Interaction (random control)	$+0.0$	$[-15.0, +15.0]$	1.000	20

Table 6. AIME 2024 paired contrasts. The direct arm is at floor (0%) in all cells, so its rows carry no information.

Every direct cell scored 0/30: at AIME difficulty Qwen3-4B cannot answer without reasoning, so the direct arm — and therefore the interaction — is uninformative by floor effect. In the CoT arm, ablation cost 11.7 points on the 20 paired problems while the random control *gained* 6.7, giving a selectivity point estimate of $+20.0$ — the largest targeted deficit observed anywhere in the study — but with a CI of $[+0.0, +40.0]$ ($p = 0.115$). The analysis tool's own pre-registered rule flags any CI wider than 25 points as unable to distinguish “no effect” from “too few problems”, and this one is 40 wide. The cell holds 20 rather than 30 problems by protocol, not accident: the degeneration gate compares ablated CoT's unusable rate to intact's over its first 20 problems, and it fired — 40% of ablated generations were unusable against intact's 5%, a $+35$ -point delta against the 15-point threshold — stopping the cell before the last 10 problems were generated. All eight unusable ablated generations ran to the 32,768-token cap, six ending in verbatim repetition loops, against one intact cap hit on the same problems. The 20-point selective deficit and the gate firing are therefore the same observation read twice: ablation's dominant effect on AIME CoT is to prevent termination.

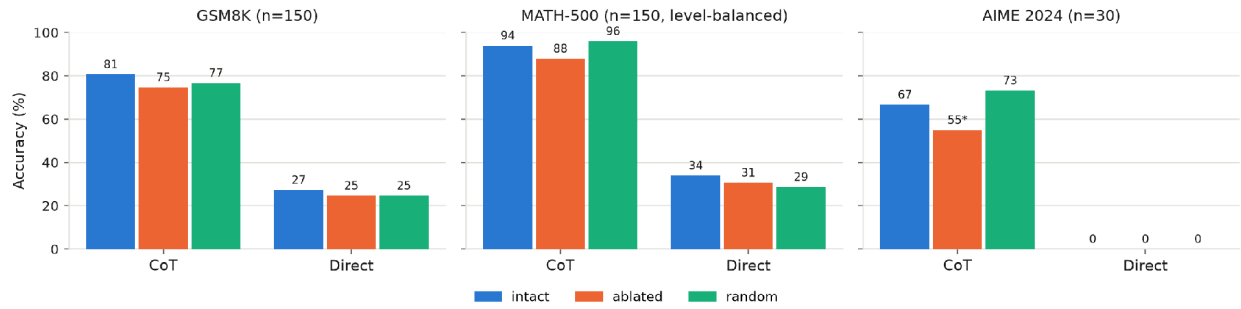


Figure 1. Accuracy by cell. *AIME cot_ablated is n = 20 of 30 (the degeneration gate fired).

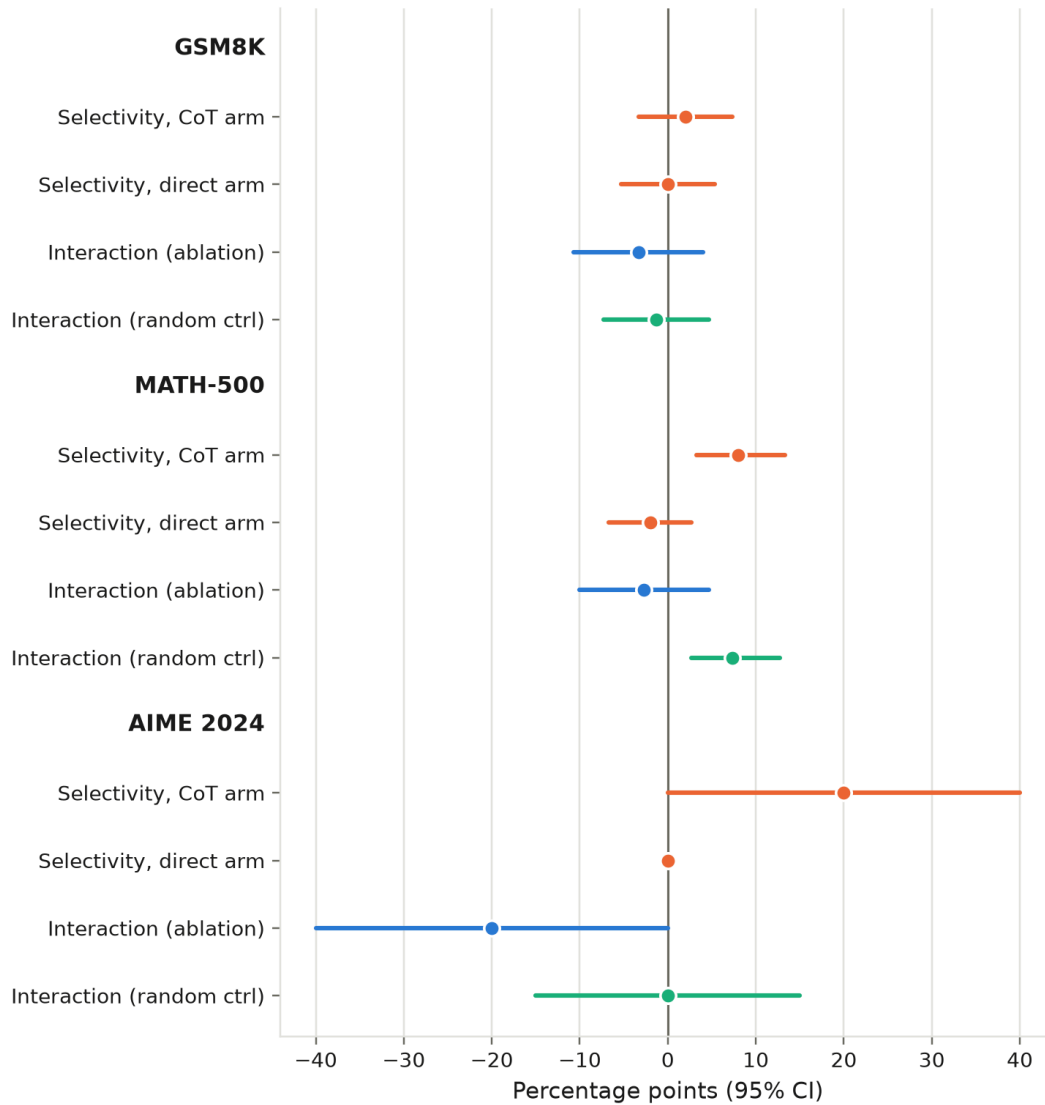


Figure 2. Pre-registered contrasts with 95% bootstrap CIs. H1 predicts a positive ablation interaction; the random-control interaction should sit at zero — on MATH-500 it does not (+7.3, $p = 0.003$). MATH-500's CoT-arm selectivity is the one contrast excluding zero.

5. Analysis of results

The substitution hypothesis is not supported. Both complete, fully-powered datasets return null interactions of the wrong sign: GSM8K at -3.3 $[-10.7, +4.0]$ and MATH-500 at -2.7 $[-10.0, +4.7]$. A substitution effect of ≥ 8 points would have been visible on either, and none was. On GSM8K the nominally significant CoT drop ($p = 0.035$) does not survive the matched control; on MATH-500, correcting for the non-zero control baseline moves the result further from H1, not closer (below). We report these as informative nulls for large substitution effects, not merely an absence of evidence.

The one significant targeted effect is in the arm H1 predicted would be protected — and it scales with difficulty. MATH-500's CoT-arm selectivity of $+8.0$ $[+3.3, +13.3]$ ($p = 0.001$) is the study's only contrast excluding zero: removing the ten J-lens directions costs CoT accuracy 8 points more than removing ten random ones. Absolute damage stays modest — the ablated model still scores 88% — and the deficit is not wrong answers but non-termination: incorrect counts fall from 3 to 1 while all 17 added failures are cap hits. AIME shows the same signature at larger magnitude but without significance ($+20.0$, CI touching zero, 40 points wide; eight of nine ablated failures at cap — the very observation that fired the gate, so the selectivity and the gate are one finding counted twice; and the direct arm's floor removes the interaction that would anchor it). Across datasets, CoT-arm selectivity is monotone in difficulty: $+2.0$ (n.s.) $\rightarrow +8.0$ ($p = 0.001$) $\rightarrow +20.0$ (underpowered). This is the complementary account's prediction (H2), not substitution's: the harder the problem, the more the model appears to lean on workspace state *while* externalizing — and the ablation's specific, selective effect is to stop chains of thought from terminating.

MATH-500's control interaction breaks the read-against-zero rule — and would have counterfeited H1. The $+7.3$ -point control interaction is in exactly the direction H1 predicts for the *ablation* interaction: had the random control not been run, an ablation interaction of that size would have read as textbook substitution. What it actually shows is that the direct arm is fragile to removing *any* ten directions in the band while CoT answering absorbs the same damage — a real internal/external asymmetry, but one about the degradation-robustness that externalized reasoning confers, not about the J-space specifically. Methodologically, it means the pre-registered rule of reading the ablation interaction against zero is insufficient on this dataset: only the difference from $+7.3$ can be attributed to J-space targeting. The regenerated cell's interaction of -2.7 sits 10.0 points *below* that baseline — exactly the direct arm's null selectivity (-2.0) minus the CoT arm's significant one ($+8.0$) — so correcting for the arm-dependent broad degradation pushes the result away from substitution, not toward it.

Alternative explanations that remain open

- **The $J = I$ approximation may miss the real J-space.** Our “J-lens” is the gain-aware logit lens, which reads what an activation is disposed to say *now*, not the paper's Jacobian-averaged disposition. A directions probe found the gold answer is not linearly readable in the ablated band at all (gold@10 = 0% through layers 14–19, first reaching 100% at layer 32), so the top-10 readout there may be tokenizer noise rather than workspace content. A null under $J = I$ does not falsify the paper's claim.
- **Scale.** The paper's clean effects are on frontier-scale models and it reports that its smallest model degraded in coherence before showing qualitative workspace effects. Qwen3-4B may simply be below the scale at which a verbalizable workspace is cleanly separable.
- **Band granularity and placement.** The paper's 16-point-deep light band maps to just 6 of 36 layers, and a fixed-width sliding-window sweep (14–19, 20–25, 26–31, plus full-width 14–33) run at the pre-registered settings found no window with selective damage — but all windows shared the band's coarseness. Strength was never escalated to the medium/heavy bands, deliberately, to avoid the incoherence confound.

- **Cap asymmetries.** Incomplete counts rise under both ablation and control (GSM8K CoT: 24 → 30/28), so part of every drop is non-termination rather than wrong reasoning. The caps are shared across states, and the control absorbs the same penalty, so this biases the selectivity estimates toward zero rather than manufacturing effects — but on AIME the 8/20 ablated cap-hit rate is what fired the gate, and on MATH-500 the significant +8.0 selectivity is carried entirely by cap hits (17 of 18 added failures). On both, non-termination is not a nuisance on top of the effect — it is the effect, and “ablation selectively removes reasoning” and “ablation selectively makes generation fail to terminate” are not yet separable.
- **Control choice.** Our control matches the operation (same k, band, projection) with randomized selection. The paper also uses matched-norm random perturbations; a norm-matched noise control (implemented but not run) could separate “removing any structured directions” from “removing this much signal”.

What I would test next

- I would re-run the 17 capped MATH-500 cot_ablated generations at a doubled cap, to separate “ablated CoT never terminates” from “ablated CoT is slower”: the +8.0 selectivity is currently carried entirely by cap censoring, and the two readings differ in what they say about the workspace.
- With more compute and time, I would raise the AIME token cap — recalibrating it with an actual measurement (run.py --calibrate-caps, up to the checkpoint's 40,960-position ceiling) instead of the uncalibrated 32,768 budget ceiling — and re-run cot_ablated under the gate. That is the defensible way to recover the cell the gate stopped: generating the missing 10 problems as-is would mean overriding a fired pre-registered gate (a documented shard script exists for it, src/shard_ablated_tail.sh, with a mandatory disclosure file), and 40% of ablated generations hitting the current cap is precisely the finding a bigger cap would let us read past.
- I would ablate different layers of Qwen3-4B. Everything here used the paper's light band mapped to layers 14–19, where the directions probe says the answer is not yet linearly readable; a dose–response curve over band width (light → medium → heavy) and a powered sweep over band position — the exploratory sliding-window comparison found no window with selective damage, but only at $n = 12$ on GSM8K — would show whether selectivity appears elsewhere in depth before coherence degrades, with the degeneration gate as the stopping rule.
- I would run the *nothink* middle rung (thinking disabled, no answer-only instruction), pre-registered but unrun — it separates “no externalization” from “forced immediate answer”.
- I would estimate actual per-layer Jacobians on a subset of prompts to test whether $J \neq I$ selects materially different directions in the workspace band — the direct test of the biggest approximation in this study.
- I would report observed power: re-run the power analysis at the measured accuracies and correlation, so the widths of the null CIs are themselves interpretable.

Reproducibility. All prompts, pre-registered constants, run manifests, and scored records are in the repository; analyze.py regenerates every number above from the committed .jsonl run files.